
Linear Representations and Equations — Telling Slope and y -Intercept Apart

Slope and y -intercept answer two different questions, and mixing them up is the single most common error in this unit.

- **Slope** m answers “how much does y change for each 1 that x increases?” It is a *rate*: $m = \frac{\text{change in } y}{\text{change in } x}$.
- **y -intercept** b answers “what is y when x is 0?” It is a single *output*, at one specific input.
- In $y = mx + b$ the slope is the number *attached to* x and the intercept is the number standing alone — no matter which one is written first.
- In each problem: name the error in words first, then show corrected work.

1. Find and fix. A line is given by $y = 3x + 5$. Lin says “the slope is 5 and the y -intercept is 3,” then finds y when $x = 4$ by computing $5(4) + 3 = 23$.

- (a) State the real slope and the real y -intercept.
(b) Find the correct value of y when $x = 4$.

Answer: _____

2. Find and fix. A line is given by $y = 12 - 2x$. Rae says “the first number is the slope, so $m = 12$ and $b = -2$,” then finds y when $x = 5$ by computing 58.

- (a) Explain why the order the numbers are written in does not decide which is which.
(b) Find the correct value of y when $x = 5$.

Answer: _____

3. Find and fix. A line passes through $(2, 3)$ and $(6, 11)$. Dev writes $m = \frac{6 - 2}{11 - 3} = \frac{1}{2}$.

- (a) Say which change Dev put on top, and which one belongs there.
(b) Find the correct slope.

Answer: _____

4. Find and fix. A linear relationship is shown in this table.

x	1	2	3	4
y	13	17	21	25

Sofia says “the rate of change is 13, because that is the first number in the y row.”

- (a) Explain what a rate of change actually measures here.
(b) Find the true change in y for each step of 1 in x .

Answer: _____

5. Find and fix. Use the same table as problem 4. Sofia now says “the y -intercept is 13, because that is where the table starts.”

- (a) Say what input the y -intercept belongs to, and why the table does not show it.
- (b) Using the step size of 4 you found in problem 4, work backwards one step and give the y -intercept.

Answer: _____

6. Find and fix. A line has slope 4 and passes through the point $(3, 20)$. Owen writes $y = 4x + 20$, “because 20 is the y -value.”

- (a) Test Owen’s equation at $x = 3$ and say what it gives.
- (b) Write the equation $4(3) + b = 20$, solve it for b , and state the correct y -intercept.

Answer: _____

7. Find and fix. A pool is draining at a steady rate. After 2 minutes it holds 46 gallons; after 8 minutes it holds 22 gallons.

A student reports “the rate of change is -0.25 gallons per minute, and the pool started with 46 gallons.”

(a) Both numbers are wrong for the same underlying reason. Find the correct rate of change in gallons per minute.

(b) Find how many gallons the pool held at 0 minutes.

Answer: _____

8. Explain.

(a) In your own words, state the question the slope answers and the question the y -intercept answers. Say why one is a rate and the other is a single value.

(b) Explain why swapping m and b in $y = mx + b$ gives a completely different line, even though the same two numbers appear.

(c) Describe how you would find the slope and the y -intercept in each of these: a table of values, a graph, and an equation with the constant written first.